

Justin Perez

Full Stack Software Engineer

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Platform and full-stack engineer who owns systems end to end — federated retrieval in Python, event-driven .NET services, and the OIDC/SSO surface of a multi-tenant SaaS platform. I write the architecture decision record, drive it to consensus, and carry it to production across nine environments.

EXPERIENCE

Software Engineer 2 · NICE

June 2025 — Present

Sandy, UT

Federated retrieval platform · CXone Expert · auth library · conversational-AI integration

- Architected a federated knowledge-retrieval service in Python from its earliest commits — owning the request/response contract, config-driven provider dispatch, and the unified contract that merges results across providers with reciprocal rank fusion. Authored four architecture decision records and drove them through review to team consensus.
- Own the platform's OIDC/SSO surface: delivered preferred-username through launch, a safe revert under incident, and a hardened re-land; hardened the multi-tenant redirector; and contributed PKCE upstream to an open-source PHP auth library before enabling it by default across the product.
- Led the migration of search analytics from a synchronous, request-blocking query onto an event-driven .NET 8 service on SNS/SQS — KEDA scale-to-zero, dead-letter handling, per-site feature-flag rollout — and authored the Terragrunt, CloudFormation, and Helm infrastructure behind it across nine environments.
- Drove a read-replica adoption program to cut primary-database load: audited ~38 services for eligibility, then moved user, tag, group, and page query paths across 7+ high-traffic tables onto replica reads.
- Designed a filter-group system — versioned, stored definitions with a management API (optimistic concurrency, soft-delete and restore, version history) — that became authoritative for provider selection and result scoping, replacing brittle per-request config with fail-loud validation.
- Closed a CWE-285 broken access-control vulnerability across the C# and PHP layers without regressing the OIDC login flow that had forced a revert of the first attempt, and fixed a prompt-injection vector found in third-party penetration testing.

Software Engineer I · Cox Automotive

June 2023 — June 2025

Draper, UT

- Increased security adoption across company software inventory from 10% to over 45% by simplifying the upload process and building more efficient defect-retrieval methods.
- Designed a graph database on NeptuneDB modeling thousands of interconnected entities — vulnerabilities, software components, teams, and users — making security data queryable across the org.
- Built tooling that generates security status reports for more than 7,500 software components.
- Built reporting ETL jobs on CloudWatch Events, ECS Fargate, and Lambda.

PROJECTS

Dynasty Futures · Production trading platform for a futures prop firm

2026

dynastyfuturesdyn.com

- Solo-built production trading platform for a futures prop firm: live account dashboards, equity and P&L curves, a payout eligibility engine, and KYC. React/TypeScript, Node/Express, Prisma, Postgres, AWS, Terraform.

Campus Core · Multi-tenant academic advising platform

2024 — Present

campus-core.com

- Multi-tenant academic advising platform backed by Microsoft for Startups; led the backend across a team of five. Isolated Postgres per university, Zoom OAuth meetings, OpenAI advising assistant. React Native/Expo, Node, Prisma, Azure.

Huddle Up · Team fantasy football platform

2023

capstone-cs.eng.utah.edu/fan-huddle

SKILLS

Languages	TypeScript, JavaScript, Python, C#, Go, PHP, Java, C++, SQL
Frontend	React, React Native, Next.js, Tailwind, HTML, CSS
Backend & data	Node.js, Express, .NET, ASP.NET Core, Prisma, PostgreSQL, MySQL, NeptuneDB, Redis, Event-driven architecture, SNS / SQS, Multi-tenancy, OpenTelemetry
Infrastructure	AWS, Azure, Terraform, Terragrunt, Docker, Kubernetes, Helm, KEDA, ECS Fargate, Lambda, GitHub Actions
Security	OIDC / SSO, OAuth2 / PKCE, JWT, Veracode, Wiz, Aqua, SAST / SCA, Vulnerability management

EDUCATION

B.S. in Computer Science · University of Utah

Salt Lake City, UT · Aug 2019 — May 2023